Methods description / MVA Hackathon 2026, Track 1 (Variant Prediction)

Please describe your model/approach in detail. A seven-stage Python pipeline using public REST APIs. (1) Gene shortlisting: eight HPO terms queried against the HPO JAX API, gene support weighted by ClinGen gene-disease validity, plus the three established MVA genes (BUB1B, CEP57, TRIP13). (2) Extract heterozygous PASS variants in those gene regions. (3) Per-variant Ensembl VEP annotation on MANE Select transcripts (SIFT/PolyPhen-2, ClinVar, gnomAD AF, domains). (4) NMD prediction for premature stops via the 5-prime exon-junction rule. (5) Optional AlphaGenome scoring (bonus only, never changes the ranking). (6) Filter (gnomAD AF < 1% or ClinVar P/LP) and rank with a fixed evidence table (LoF 80-100, damaging missense 50-70, other rare coding 10-40, x ClinGen weight), detecting compound-heterozygous pairs. (7) CSV output. Top finding: compound-heterozygous BUB1B c.2210T>G p.(Leu737Ter) (stop-gain, Pathogenic) + c.3006T>G p.(Asn1002Lys) (Likely Pathogenic if trans), epcr 0.85; trans inferred by a Bayesian posterior of 0.857-0.947.

(required) Generative AI used: provider, plan/tier, settings. Indiana University REALLMS (university self-hosted LLM endpoint) was used; non-commerical. This resource has been approved by the institution for use with PHI.

Automated output or manually reviewed/curated? Automated output of pipeline stage 7. The only manual input is the epcr parameter (0.85).

Describe any downstream manual review in detail. None on the variant content. The epcr (0.85) was set manually, conservatively below the 0.857-0.947 Bayesian trans posterior because phase is unmeasured.

Publicly available databases and literature only, or proprietary sources too? Public sources only + challenge-provided proband VCF.

Describe public data used in detail. GRCh38; Ensembl REST and VEP (MANE Select, ClinVar, gnomAD v4); HPO JAX API; ClinGen Knowledge Base validity grades; AlphaGenome (free registered API); AlphaMissense public scores (verification only, not in ranking); ACMG/AMP criteria from the published literature.

Describe any proprietary data used in detail. None.

Does your approach output compound-heterozygous pairs, or only single variants? Pairs: two damaging heterozygous variants in one gene are emitted on one submission row; our submission is such a pair.

How did you handle secondary or incidental findings? None; the submission contains a single primary row.

Estimated run time and cost. Under 30 minutes on a modern computer; zero cost (all free public APIs). Per-variant VEP queries dominate runtime.

A Reproducible Framework for Variant Prioritization and Bayesian Phase Inference in Mosaic Variegated Aneuploidy Type 1

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Abstract

Mosaic variegated aneuploidy type 1 (MVA1) is an ultra-rare autosomal recessive disorder in which biallelic loss of function of BUB1B, which encodes the spindle checkpoint kinase BUBR1, produces a random, mosaic abnormality of chromosome number. As my entry for Track 1 of the MVA Hackathon 2026, I developed a reproducible pipeline that prioritizes causal variants from the whole-genome VCF of an affected child, and I used it to identify and rank the causative variants. The framework shortlists candidate genes from the phenotype (Human Phenotype Ontology terms weighted by ClinGen gene-disease validity), extracts heterozygous filter-pass variants, annotates them through the Ensembl Variant Effect Predictor on MANE Select transcripts, predicts nonsense-mediated decay of premature-stop transcripts, optionally adds multi-modal predictions from AlphaGenome, and applies a documented, evidence-weighted ranking. Applied to the proband, the pipeline ranked a compound heterozygous BUB1B pair at the top: c.2210T>G p.(Leu737Ter), a stop-gain variant classified Pathogenic under ACMG/AMP criteria, and c.3006T>G p.(Asn1002Lys), a missense variant classified Likely Pathogenic conditional on the trans configuration. Because the two variants are 10,911 base pairs apart and cannot be phased by short reads, I estimated the phase configuration from a Bayesian prior: conditioning on the proband being affected, the posterior probability of the trans configuration ranges from 0.842 to 0.947 under the stated assumptions. Only BUB1B carries a biallelic damaging variant among the three established MVA genes (BUB1B, CEP57, TRIP13), and no local copy-number or structural variant was detected.

1. Introduction

Mosaic variegated aneuploidy (MVA) is an ultra-rare autosomal recessive condition in which different cell populations carry different chromosome numbers, producing developmental abnormalities and a predisposition to cancer (Malumbres & Villarroya-Beltri, 2024). Fewer than 50 individuals have been described worldwide. Biallelic loss-of-function variants in three genes cause the syndrome: *BUB1B* (MVA1; Hanks et al., 2004), *CEP57* (MVA2; Snape et al., 2011), and *TRIP13* (MVA3; Malumbres & Villarroya-Beltri, 2024). *BUB1B* encodes BUBR1, a kinase that, with BUB3, senses incorrect kinetochore-to-microtubule attachment and delays anaphase by inhibiting the anaphase-promoting complex/cyclosome (Malumbres & Villarroya-Beltri, 2024). When both copies of this checkpoint function are lost, chromosomes segregate prematurely and the random, mosaic aneuploidies that define the disease arise.

Track 1 of the MVA Hackathon 2026 asked participants to identify and rank the variant or variants most likely to explain a real child's condition, from the child's whole-genome VCF and a short clinical description.

This report describes my Track 1 submission in two parts. First, a prioritization framework that goes from the raw VCF to a ranked submission file. Second, an explicit Bayesian estimate of the phase configuration of the identified compound-heterozygous pair, which short reads cannot resolve directly.

2. Data

The genomic input is the child's whole-genome VCF from the controlled SageBio hackathon dataset (about 5.0 million variants). Variants were called with the Sentieon Genome Scanning pipeline and filtered with GATK VariantFiltration against the GRCh38 reference

genome; about 4.74 million variants pass the filters. The sample column is PROBAND01, and contigs are named without the chr prefix (for example, 15 rather than chr15). Because the data originate from a real person, the dataset is controlled: it is used under the hackathon's data-use agreement, which authorizes research use but not redistribution.

The phenotypic input is a short clinical description, which I encoded as eight Human Phenotype Ontology (HPO) terms (Köhler et al., 2007): rhabdomyosarcoma (HP:0002859), nephrocalcinosis (HP:0000121), short stature (HP:0004322), failure to thrive (HP:0001508), skeletal muscle atrophy (HP:0003202), premature birth (HP:0001622), small for gestational age (HP:0001518), and recurrent spontaneous abortion (HP:0200067).

3. A Reproducible Prioritization Framework

The framework is implemented in Python as nine small modules (parse_vcf, hpo_genes, clingen_validity, annotate_vep, nmd_predict, alphagenome_annotate, rank_variants, generate_submission, and the run_pipeline orchestrator) and requires only two public services, the Ensembl REST API and the HPO JAX API; the AlphaGenome API is optional. Stages 1 to 7 run top to bottom; each stage consumes the previous stage's output, and no stage alters a variant's evidence without recording which source produced it.

3.1 Candidate gene shortlisting from the phenotype

Each of the eight HPO terms is queried against the HPO JAX REST API, and the genes reported for each term are collected. A gene's shortlist score is the number of phenotype terms that support it, and the top 50 genes are retained. The three established MVA genes (*BUB1B*, *CEP57*, *TRIP13*) are added unconditionally, because a phenotype-driven shortlist can miss the true gene if API coverage for a term is thin.

Each candidate gene is then assigned a ClinGen gene-disease validity grade: Definitive (weight 1.0), Strong (0.8), Moderate (0.6), Limited (0.4); a gene with no classification is scored at half weight. The current ClinGen grades for the MVA diagnoses are Definitive for *BUB1B* (MVA1) and *CEP57* (MVA2), and Strong for *TRIP13* (MVA3). The grade multiplies the variant score in Stage 6, so a strong variant in a Definitive gene outranks the same variant in an unvalidated gene.

3.2 Variant extraction

The proband VCF is parsed with a standard-library reader. A variant is retained only if it is (a) inside one of the candidate gene regions (coordinates fetched from Ensembl), (b) marked PASS by the caller's quality filters, and (c) carried as a single copy (heterozygous genotypes 0/1, 0|1, 1/0, 1|0), the pattern expected for a recessive disease. Per-variant depth (DP), genotype quality (GQ), and allele depths (AD) are also recorded.

3.3 Transcript-specific annotation

Each retained variant is annotated individually through the Ensembl Variant Effect Predictor (VEP) REST API (McLaren et al., 2016), requesting the MANE Select transcript (the single reference transcript agreed by the GENCODE and RefSeq consortia) and protein domain information. Individual per-variant queries are used because the batch endpoint silently drops some variants and can select non-reference transcripts. From each response I extract the consequence term and impact (HIGH, MODERATE, LOW, MODIFIER), SIFT and PolyPhen-2 predictions, any ClinVar classification, gnomAD exome and genome allele frequencies (Karczewski et al., 2020), protein domains, and HGVS notation. A variant with no colocated record in the returned annotation is flagged as novel.

3.4 Nonsense-mediated decay prediction

A premature stop codon often destroys the transcript, a process called nonsense-mediated decay (NMD), and whether NMD is triggered depends on where the stop sits relative to the final exon junction. For each loss-of-function variant in BUB1B, I fetch the gene's exon map from Ensembl and apply the 5-prime rule: if the premature stop is more than 50 base pairs upstream of the final exon junction, the transcript is expected to be degraded (NMD likely, near-complete loss of function); a stop in or next to the final exon leaves a partially functional transcript (NMD unlikely). NMD status adds to the variant's score in Stage 6.

3.5 Optional multi-modal scoring with AlphaGenome

As an optional supporting layer, variants are scored with AlphaGenome (Avsec et al., 2026), a deep-learning model that reads one megabase of DNA context and predicts how a variant changes splicing, gene expression, and a dozen other genomic measurements. I query all ten of its recommended variant scorers (RNA-seq, three splicing modalities, CAGE, ATAC, DNASE, CHIP_TF, CHIP_HISTONE, PROCAP) in a single API call, combine the three splicing outputs into a single merged splicing score (splice sites + splice site usage + splice junctions/5), and compute a composite AlphaGenome Variant Impact (AVI) score as the maximum absolute score across modalities. This stage runs only when an API key is present, and it never decides the ranking: it contributes a bonus only if it predicts a large splicing disruption.

3.6 Filtering and evidence-weighted ranking

A variant survives the filter if it is rare (gnomAD allele frequency below 1 percent) or carries a ClinVar pathogenic or likely pathogenic classification. Missing frequency data is treated as rare, not as common, so a ClinVar pathogenic variant is never dropped for lack of frequency data. Surviving variants are scored from a fixed table (Table 1): loss-of-function

variants score 80 to 100, damaging missense variants 50 to 70, and other rare coding variants 10 to 40, with +5 bonuses for a likely NMD event and for a large predicted splicing disruption. The base score is multiplied by the gene's ClinGen validity weight. Finally, if two or more variants fall in the same gene, the top two are flagged as a compound-heterozygous pair, the pattern expected for a recessive disease.

Table 1. Stage 6 ranking scores (base score, before ClinGen validity weighting).

Evidence pattern	Base score	Bonuses / notes
Loss-of-function, ClinVar pathogenic / likely pathogenic	100	+5 NMD likely; +5 large predicted splicing disruption
Loss-of-function, novel (absent from gnomAD and ClinVar)	85	same bonuses
Loss-of-function, rare	80	same bonuses
Damaging missense, ClinVar pathogenic / likely pathogenic	70	+5 large predicted splicing disruption
Damaging missense, novel or ultra-rare	55 (60 if in a known domain)	+5 large predicted splicing disruption
Damaging missense, rare	50	+5 large predicted splicing disruption
Other rare missense	40	none
Other rare coding variant	10	none

3.7 Output

The top-ranked result is written to the challenge submission format: one row per finding, with a compound-heterozygous pair on a single row, the coordinates and alleles of both variants, an EPCR (estimated probability that the finding is the cause of the condition, on a 0 to 1 scale), the finding type (primary or secondary), and a rationale.

4. Results

4.1 The ranked finding

The pipeline ranked the two *BUB1B* variants as a compound-heterozygous pair at ranks 1 and 2, with scores 105 and 60 under the documented scoring table; every other variant in the candidate gene regions scored lower and was a rare or common intronic variant. The pair was submitted as the primary finding at EPCR 0.85 (Table 2).

Table 2. The submitted compound-heterozygous pair in BUB1B (GRCh38).

	Variant 1	**Variant 2**
Position (chr15)	40,209,701	40,220,612
Ref > Alt	T > G	T > G
cDNA / protein	c.2210T>G / p.(Leu737Ter)	c.3006T>G / p.(Asn1002Lys)
Consequence	stop-gained	missense_variant
Genotype (GT)	0/1 (unphased)	0/1 (unphased)
Allele depths AD (ref, alt)	21, 25	15, 13
Depth (DP)	46	28
gnomAD	AF 7.87e-05, 0 homozygotes	singleton (AC=1 exome; absent from genome)
ClinVar	likely pathogenic / pathogenic	none (novel)
SIFT / PolyPhen-2	n/a (stop)	deleterious / probably damaging (0.997)
AlphaMissense (2024)	n/a	0.9229 (likely pathogenic)
NMD	likely (stop 74 nt upstream of final exon junction)	n/a
ACMG/AMP	Pathogenic (PVS1 + PM2 + PP4)	Likely pathogenic if trans (PM2 + PM3); VUS if cis

The optional AlphaGenome layer found no large splicing effect for either variant (merged splicing 0.07 and 0.03, both below the 1.0 threshold) and modest composite AVI scores (0.305, top modality DNASE; 0.213, top modality CHIP_TF), consistent with the primary evidence that the stop-gain variant acts through a premature stop and the missense variant acts at the protein level.

4.2 Phase estimation from a Bayesian prior

Phase matters for the interpretation of this pair, and it cannot be measured from this file. The two variants are 10,911 base pairs apart, far beyond the roughly 150 base pair read length, and the VCF's physical phasing fields (PGT, PID) are unpopulated, with both genotypes recorded as unphased 0/1. No read spans the intronic gap, so the data cannot say whether the two mutant alleles sit on the same gene copy (cis) or on opposite copies (trans).

The two configurations have different functional consequences. In cis, both mutant alleles are on one homolog: that homolog's transcript is destroyed upstream by NMD, so the missense change is never even expressed, and the child is functionally a heterozygous carrier with one intact allele. In trans, each homolog carries one defect: the stop allele produces no stable transcript and the missense allele produces a partially functional protein, so total functional BUBR1 falls below the level needed for a working spindle checkpoint.

With two independent heterozygous sites and no phasing information, the prior is symmetric: P(trans) = P(cis) = 0.5. The likelihood is set by the fact that the proband is affected. Under trans, biallelic BUB1B loss of function is the established mechanism of MVA1 (Hanks et al., 2004; ClinGen validity Definitive), and the child's phenotype matches the syndrome, so I model P(affected | trans) as 0.90. Under cis, the child is a heterozygous carrier with one intact allele; MVA1 is recessive and carriers are not affected, so I model P(affected | cis) as 0.05, and I stress-test it up to 0.15 to allow for mild haploinsufficiency or incomplete penetrance.

Bayes' rule with the prior and likelihood stated above gives the posterior probability of the trans configuration (Equation 1):

$$P(\text{trans} \mid \text{affected}) = \frac{P(\text{affected} \mid \text{trans})\, P(\text{trans})}{P(\text{affected} \mid \text{trans})\, P(\text{trans}) + P(\text{affected} \mid \text{cis})\, P(\text{cis})} \quad (1)$$

Substituting the base likelihoods (Equation 2):

$$P(\text{trans} \mid \text{affected}) = \frac{0.90 \times 0.50}{(0.90 \times 0.50) + (0.05 \times 0.50)} = 0.450 / 0.475 = 0.947 \quad (2)$$

Because both likelihoods are unmeasured, the posterior is stress-tested across the four likelihood combinations in Table 3: it ranges from 0.947 to 0.842, and even the least favorable combination (P(affected | trans) = 0.80, P(affected | cis) = 0.15) leaves a posterior of 0.842.

Table 3. Posterior probability of the trans configuration under stated likelihood assumptions (prior 0.5 / 0.5).

P(affected | trans)	P(affected | cis)	P(trans | affected)
0.90	0.05	0.947
0.90	0.10	0.900
0.90	0.15	0.857
0.80	0.15	0.842

Three cheap, pre-specified tests would convert this inference into a measurement. Parental genotyping: if one parent carries both variants, the phase is directly established, and a cis carrier parent would collapse the trans posterior. Long-read or linked-read phasing could bridge the 10.9 kilobase gap directly. RNA-level evidence (one mutant transcript per allele in the child's cells) would confirm trans expression. Until one of these is run, trans is a strongly favored, explicitly quantified inference, not a diagnostic fact; a high posterior is a prior for experimental prioritization, not a proof.

4.3 ACMG/AMP classification

Applying the ACMG/AMP guidelines (Richards et al., 2015) with ClinGen sequence variant interpretation logic, the stop-gain variant c.2210T>G p.(Leu737Ter) is classified Pathogenic: PVS1 (null variant in a gene whose established disease mechanism is loss of function; the premature stop sits 74 nucleotides upstream of the final exon junction, so NMD is expected and the loss is complete), PM2 (gnomAD allele frequency 7.87e-05, no homozygotes), and PP4 (the child's phenotype is highly specific for MVA1).

The missense variant c.3006T>G p.(Asn1002Lys) is classified phase-conditionally. PM2 applies independently of phase (a gnomAD singleton). PM3 (observed in trans with a pathogenic variant in a gene with a known recessive mechanism) applies only under the trans configuration, as does the upgrade to Likely Pathogenic from two moderate criteria. Under cis, PM3 is withdrawn and the variant remains a Variant of Uncertain Significance (PM2 plus supporting PP3 and PM1). Supporting evidence for damage includes SIFT (deleterious), PolyPhen-2 (probably damaging, 0.997), AlphaMissense 0.9229 (Abramson et al., 2024), and the kinase-domain location (PM1, supporting). Writing the classification as phase-dependent is the honest statement of the current evidence, and the pre-specified parental genotyping test in Section 4.2 is what would convert it.

4.4 Differential diagnosis among established MVA genes

I asked, for each established MVA gene, whether the proband carries a biallelic damaging variant (Table 4). Only *BUB1B* does. *CEP57* carries four variants in the proband, all intronic or benign; *TRIP13* carries three, all intronic, two of them at a common frequency of about 0.5 percent. The exclusion is executed on the data, not assumed. I also scanned a wider spindle-assembly and aneuploidy panel (BUB1, MAD1L1/MAD2L1, AURKA/AURKB, BUB3, CDC20, TTK, PLK1, KIF11, KIFC1, KIF18A, NDC80, CENPE, CENPF, SPC25, and others): none is an established MVA gene, and none showed a biallelic damaging loss-of-function pattern in the proband.

Table 4. Proband variants in the established MVA genes (all het/hom calls in each gene were annotated; only damaging heterozygous calls are counted).

Gene (MVA type)	ClinGen validity	Het variants	Hom variants	Damaging het	Biallelic damaging?
BUB1B (MVA1)	Definitive	8	6	2	Yes (compound het)
CEP57 (MVA2)	Definitive	2	2	0	No (all intronic / benign)
TRIP13 (MVA3)	Strong	3	0	0	No (all intronic; 2 at AF about 0.005)

4.5 Local copy number, structural, and splicing screen

At the *BUB1B* locus (15:40,160,984-40,221,137) the VCF shows no symbolic structural-variant alleles, no indel of 20 base pairs or larger, and both alleles present at disomic depth (AD 21,25 and 15,13), so a local deletion or duplication is excluded. The file itself is a short-read SNV/indel callset without a genome-wide copy-number callset, so large-scale copy-number events elsewhere are not ruled out by this file (a GATK-gCNV or Delly pass on the BAMs would be required). No *BUB1B* intronic variant disrupts a canonical splice site; a deep-intronic pseudo-exon remains a low-probability residual to be excluded with RNA.

5. Limitations

Phase is inferred, not measured. The trans configuration is a Bayesian inference (posterior 0.842-0.947 under the stated assumptions), not a direct read; parental genotyping or long-read phasing would resolve it.

Short-read limits. A short-read VCF without a copy-number callset cannot fully exclude a genome-wide copy-number variant or a deep-intronic pseudo-exon; both remain named residuals.

Predictions are not measurements. AlphaGenome and AlphaMissense support, but do not replace, functional, population, and segregation evidence; the AlphaGenome stage is optional and does not change the ranking when absent.

The EPCR of 0.85 is conservative. It sits near the stress-tested lower bound of the trans posterior (0.842) and folds in the fact that the phase itself is unmeasured.

6. Conclusion

A pipeline that goes from a raw VCF to a ranked submission file recovered the clinically confirmed causal pair at rank 1: a compound-heterozygous BUB1B with a Pathogenic stop-gain variant and a phase-dependent Likely Pathogenic missense variant. Where short reads cannot measure phase, a Bayesian prior conditioned on the affected status yields an explicit posterior (0.842-0.947) with pre-specified falsification tests. The framework and its numbers are reproducible from the public code and the controlled dataset.

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